

Computational Repurposing Screen of FDA-Approved Anti-Inflammatory and Antiviral Drugs Against BACE1 (β -Secretase 1): A Blind Molecular Docking Study

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ABSTRACT

Beta-secretase 1 (BACE1) is the rate-limiting enzyme in amyloid-beta production and a genetically validated drug target for Alzheimer's disease, yet every purpose-built BACE1 inhibitor to reach Phase 2/3 clinical trials (verubecestat, lanabecestat, atabecestat, umibecestat, and elenbecestat) has failed on grounds of futility, hepatotoxicity, or cognitive worsening. This creates a rationale for exploring drug repurposing rather than de novo inhibitor design. This study screened 36 FDA-approved compounds from two mechanistically underexplored classes — anti-inflammatory drugs (n = 20) and antivirals (n = 16) — against a high-confidence AlphaFold structure of human BACE1 (UniProt P56817; pLDDT 87.48) using blind molecular docking (CB-Dock2 / AutoDock Vina). Active-site residues were identified from the literature (catalytic dyad Asp32/Asp228 in canonical numbering, corresponding to Asp93/Asp289 in the full-length AlphaFold model) and the docking method was validated by reproducing a literature Vina score for the clinical BACE1 inhibitor atabecestat (−9.61 kcal/mol against PDB 1FKN). All 36 compounds were independently redocked to a single dominant cavity (1217 Å³) whose contact-residue list included both catalytic aspartates in every case, supporting the validity of the blind-docking protocol. The six HIV-1 protease inhibitors in the library occupied five of the top eight ranks overall (lopinavir −10.6, saquinavir −10.1, darunavir −9.6, nelfinavir −9.5, atazanavir −9.2, and ritonavir −8.9 kcal/mol), each meeting or approaching the atabecestat benchmark — a result consistent with the shared aspartic-protease catalytic mechanism of BACE1 and HIV-1 protease. A secondary, weaker signal was observed among heteroaromatic/sulfonamide-containing anti-inflammatory drugs (celecoxib, meloxicam, piroxicam, baricitinib, sulfasalazine), while small, polar, or rigid nonpolar compounds (nucleoside analogs, cage amines) consistently scored below −7.0 kcal/mol. These findings nominate HIV-1 protease inhibitors as a structurally rationalized, testable hypothesis for BACE1-directed drug repurposing, while underscoring that docking scores are hypothesis-generating rather than proof of binding, pending enzymatic and structural validation.

INTRODUCTION

Alzheimer's disease (AD) is the most common cause of dementia and one of the largest unmet medical needs in modern medicine. In the United States alone, an estimated 7.2 million people aged 65 and older are living with Alzheimer's dementia, a figure projected to reach 13.8 million by 2060, with annual care costs projected at \$384 billion in 2025 [1]. Globally, more than 55 million people live with dementia, a number projected to reach 139 million by 2050, with an estimated global economic cost of \$1.3 trillion [2].

The amyloid cascade hypothesis holds that the accumulation of amyloid-beta ($A\beta$) peptide is the central, initiating event in AD pathogenesis [3, 4]. $A\beta$ is generated when amyloid precursor protein (APP) is sequentially cleaved by two enzymes: β -secretase, which performs the initial, rate-limiting cut, and γ -secretase, which completes the process. β -site APP-cleaving enzyme 1 (BACE1) was identified in 1999 as the primary β -secretase [5], and BACE1 knockout mice fail to generate $A\beta$ while otherwise appearing grossly normal in early studies [6], establishing BACE1 as a genetically validated, mechanistically central drug target. BACE1 is a pepsin-family transmembrane aspartic protease whose catalytic domain contains two DTGS/DSGT signature motifs that together form the active site [7].

Despite this strong mechanistic rationale, every BACE1 inhibitor to reach late-stage clinical trials has failed. Verubecestat lowered cerebrospinal fluid $A\beta$ by 63–81% yet produced no slowing of cognitive decline and was associated with worsening cognition at higher doses [8, 9]; lanabecestat was discontinued for futility [10]; atabecestat was halted for hepatotoxicity and dose-related cognitive worsening [11]; and umibecestat and elenbecestat were both discontinued after interim analyses showed cognitive worsening or an unfavorable risk–benefit profile [12]. This pattern of failure — potent target engagement without clinical benefit, and in several cases outright harm — has led some researchers to argue that the field's high-potency, fully-blocking inhibitors were mechanistically the wrong approach, and that lower-magnitude or differently-timed BACE1 modulation might be safer and more effective [13]. This unmet need and mechanistic uncertainty motivate the search for alternative, already-approved compounds that engage the BACE1 active site through different chemotypes than the failed clinical candidates.

Two independent lines of evidence motivate screening anti-inflammatory and antiviral drug classes specifically against BACE1. First, a substantial epidemiological literature associates long-term nonsteroidal anti-inflammatory drug (NSAID) use with reduced AD risk: the Rotterdam Study found a relative risk of 0.20 for AD among long-term (>24 month) NSAID users [14], and a pooled analysis across six cohorts found an adjusted hazard ratio of 0.77 [15]. This association is not without contradiction — the randomized ADAPT prevention trial of naproxen and celecoxib was halted early for cardiovascular safety and did not demonstrate a prevention benefit, though extended follow-up suggested a possible protective effect when NSAIDs were given years before symptom onset [16, 17] — but the overall observational signal, combined with NSAIDs' distinct chemotypes from failed BACE1 candidates, makes this drug class worth testing directly against the BACE1 active site.

Second, BACE1 belongs to the aspartic protease family, which also includes renin, the plasmepsins, and HIV-1 protease [18]. Because members of this family share a conserved catalytic mechanism — two active-site aspartate residues that activate a water molecule for nucleophilic attack on the substrate's scissile peptide bond — transition-state-mimicking inhibitor scaffolds developed against one family member have historically shown activity against others; BACE1 inhibitor design itself has explicitly drawn on lessons from HIV protease and renin inhibitor chemistry [18, 19]. Direct precedent exists: HIV-1 protease inhibitors ritonavir and lopinavir have been co-crystallized with and shown to inhibit the malarial aspartic proteases plasmepsin II and X [20], and a 2025 multitarget virtual screen of 3,468 approved drugs against Alzheimer's disease targets, including BACE1, identified the HIV protease inhibitor saquinavir as a top-ranking binder [21]. This cross-family mechanistic plausibility motivates including HIV-1 protease inhibitors as the antiviral arm of this screen's compound library, alongside nucleoside/nucleotide analogs and other antiviral chemotypes for comparison.

Drug repurposing — identifying new therapeutic uses for existing, already-approved compounds — offers a substantially faster and cheaper path to a clinical candidate than de novo drug discovery, since repurposed compounds already have established human safety and pharmacokinetic profiles [22]. Molecular docking is a standard first-pass computational triage tool for such screens, estimating the geometric and energetic complementarity between a small molecule and a protein binding pocket. This study used an AlphaFold-predicted structure of human BACE1 together with blind molecular docking (CB-Dock2, built on AutoDock Vina) to screen

a 36-compound library of FDA-approved anti-inflammatory and antiviral drugs against the BACE1 active site, with the specific aims of: (1) identifying and validating the correct binding pocket using literature-derived active-site residues and a re-docking benchmark against the known BACE1 inhibitor atabecestat; (2) ranking all 36 compounds by predicted binding affinity; and (3) evaluating whether either underexplored drug class — and HIV-1 protease inhibitors in particular — shows a structurally rationalized signal worth pursuing experimentally.

MATERIALS AND METHODS

Target Identification and Structure Preparation

Human BACE1 (UniProt accession P56817) was selected as the docking target. A predicted structure was retrieved from the AlphaFold Protein Structure Database (entry AF-P56817-F1), spanning all 501 residues of the full-length precursor with a global per-residue confidence (pLDDT) of 87.48, indicating high-confidence backbone and side-chain placement across the modeled region [27, 28]. The full-length AlphaFold numbering (beginning at Met1 of the signal peptide) was used throughout structure preparation and docking.

Active-Site Identification

Active-site residues were identified from the published BACE1 structural biology literature rather than *de novo*, since BACE1's catalytic mechanism and substrate-binding subpockets are extensively characterized. BACE1 is an aspartic protease whose activity depends on a catalytic dyad of two aspartate residues; in the numbering conventions used by most crystallographic studies (built on the mature, processed enzyme, e.g., PDB 1FKN), these are Asp32 and Asp228. Cross-referencing two independent literature sources that reported this dyad in both classic and full-precursor numbering established a consistent +61-residue offset between the two schemes, placing the catalytic dyad at Asp93 and Asp289 in the full-length UniProt/AlphaFold numbering used in this study. Additional literature-derived active-site features used to validate docking poses included the flexible “flap” loop (Tyr71/Thr72/Gln73 → Tyr132/Thr133/Gln134 in AlphaFold numbering) and the S1, S1', and S2 substrate-binding subpockets, summarized in Table 1.

Compound Library Construction

A library of 36 FDA-approved compounds was assembled from two mechanistically underexplored drug classes relative to prior BACE1 repurposing work: 20 anti-inflammatory agents (NSAIDs, disease-modifying antirheumatic drugs, corticosteroids, and targeted anti-inflammatory small molecules) and 16 antivirals (nucleoside/nucleotide analogs, an M2 ion channel blocker class, a neuraminidase inhibitor, and six HIV-1 protease inhibitors). SMILES structures for each compound were compiled and converted to three-dimensional geometries using RDKit (ETKDGv3 embedding followed by MMFF94 energy minimization). As a structural quality-control step, the molecular formula and molecular weight computed from each SMILES string were cross-checked against literature/database values for the corresponding real drug. This process identified 7 of the initial 36 SMILES strings as structurally incorrect (sulindac, sulfasalazine, leflunomide, apremilast, montelukast, lopinavir, and saquinavir) — each was syntactically valid but described the wrong molecule, an error that a syntax check alone would not have caught. Corrected structures were re-sourced from primary chemical databases and re-verified before proceeding; all 36 final structures matched their expected molecular formula and weight.

Molecular Docking Protocol

Blind molecular docking was performed using CB-Dock2, a web-based docking platform that combines automated cavity detection (CurPocket) with AutoDock Vina scoring and, where applicable, homologous-template pose refinement [26]. For each of the 36 ligands, the AlphaFold BACE1 structure was submitted as the receptor without a predefined binding site, allowing CB-Dock2 to identify candidate cavities de novo. CB-Dock2 returned, for each compound, up to five candidate binding cavities ranked by predicted Vina binding affinity (kcal/mol; more negative indicates stronger predicted binding), along with cavity volume, docking box center and dimensions, and the full list of protein residues contacting the docked pose in each cavity. The best-scoring cavity for each compound was recorded as its primary result.

Method Validation

Two validation checks were applied before interpreting the screening results. First, docking scores were benchmarked against a literature AutoDock Vina score for the known clinical BACE1 inhibitor atabecestat, which achieved -9.61 kcal/mol when docked against the BACE1 crystal structure PDB 1FKN [23], providing a

reference point for what score range corresponds to a clinically advanced BACE1 inhibitor. Second, the identity of the top-ranked cavity was cross-checked against the literature-derived active-site residues in Table 1: a docking result was considered to have engaged the genuine catalytic site only if its contact-residue list included both catalytic aspartates (Asp93 and Asp289).

RESULTS

Active-Site Validation

Across all 36 compounds, the highest-scoring cavity was, without exception, a single 1217 Å³ pocket centered at approximately (−12.0, −3.9, −4.7). This pocket's contact-residue list included both catalytic aspartates (Asp93 and Asp289) for every one of the 36 docked compounds, along with the flap residues (Tyr132, Thr133, Gln134) and S1/S1' pocket residues (Leu91, Phe169, Gly95) identified from the literature (Table 1). Because CB-Dock2 performs de novo cavity detection independently for each ligand — with no fixed search box imposed across runs — this consistent convergence on a single, catalytically defined cavity across 36 chemically diverse compounds is strong internal evidence that the docking protocol correctly identifies BACE1's genuine active site rather than an arbitrary surface pocket. This is illustrated concretely by the largest cavity detected in the structure (2163 Å³), which lies on a distinct, non-catalytic region of the protein (residues 21–30 and 315–380) and scored consistently worse than the smaller, catalytically-relevant pocket across every compound tested — despite being nearly twice the volume, confirming that cavity size alone does not predict docking success in this system.

The re-docked atabecestat benchmark score of −9.61 kcal/mol against PDB 1FKN [23] was used as the reference threshold for a screening “hit”: a compound scoring at or better than approximately −9 kcal/mol was considered to be in the same range as a clinically advanced BACE1 inhibitor.

Full Screening Results

All 36 compounds were successfully docked to the same catalytic-site cavity. Table 2 presents the complete ranked results.

HIV-1 Protease Inhibitors Occupy the Top of the Ranking

The single most consistent finding of this screen was the performance of the six HIV-1 protease inhibitors included in the antiviral sublibrary. These six compounds occupied five of the top eight overall ranks: lopinavir (−10.6 kcal/mol, rank 1), saquinavir (−10.1, rank 2), darunavir (−9.6, rank 3), nelfinavir (−9.5, rank 4), atazanavir (−9.2, rank 6), and ritonavir (−8.9, rank 8, tied). Every one of the six met or exceeded the atabecestat benchmark score of −9.61 kcal/mol, or came within 0.7 kcal/mol of it. No other structural class in the 36-compound library showed comparable internal consistency: the six protease inhibitors spanned a narrow 1.7 kcal/mol range (−10.6 to −8.9) despite representing six chemically distinct FDA-approved drugs, developed independently over more than a decade against a different target (HIV-1 protease). Contact-residue analysis showed these compounds engaging an unusually large footprint of the pocket (30–45 residues, versus 15–25 for smaller compounds), consistent with their large, peptidomimetic scaffolds making extensive simultaneous contact across the S1, S1', S2, and flap subpockets.

Secondary Findings Among Anti-Inflammatory Compounds

Among the 20 anti-inflammatory compounds, montelukast (−9.0 kcal/mol) was the single strongest performer in the entire dataset outside the protease inhibitor class, exceeding the atabecestat benchmark despite lacking the peptidomimetic architecture that plausibly explains the protease-inhibitor results. A second, more diffuse pattern was observed among heteroaromatic and sulfonamide-containing anti-inflammatory compounds: baricitinib (−8.9), celecoxib (−8.8), methotrexate (−8.6), piroxicam (−8.5), sulindac (−8.5), dexamethasone (−8.5), sulfasalazine (−8.4), prednisone (−8.4), meloxicam (−8.3), and apremilast (−8.3) formed a tightly clustered group spanning multiple distinct mechanistic classes (JAK inhibitors, COX inhibitors, a DMARD, corticosteroids, and a PDE4 inhibitor), suggesting that several distinct structural strategies — not one shared pharmacological mechanism — can achieve favorable shape and polarity complementarity with the BACE1 active site.

A controlled within-class comparison was possible among the eight NSAIDs screened, which spanned a 2.6 kcal/mol range: celecoxib (−8.8) > piroxicam $\approx$ sulindac (−8.5) > meloxicam (−8.3) > indomethacin (−7.9) > diclofenac $\approx$ ketoprofen (−7.6) > naproxen (−7.5) > ibuprofen (−6.6) > aspirin (−6.2). This gradient tracked

structural complexity and heteroaromatic/sulfonamide content rather than pharmacological subclass: celecoxib, piroxicam, sulindac, and meloxicam — the four strongest NSAID scorers — each carry a sulfone, sulfonamide, or thiazine/oxicam heteroaromatic system, while aspirin, the weakest scorer, is a small, simple acetylated benzoic acid ester with no such features.

Weakest-Performing Compounds

The bottom of the ranking was dominated by two structural groups: small polar nucleoside analogs (acyclovir -6.3, ribavirin -6.3, lamivudine -6.1) and rigid cage amines (rimantadine -6.3, amantadine -6.1). No compound in either group cleared -7.0 kcal/mol, a consistent negative result across chemically distinct scaffolds sharing only small size and either high polarity or rigid nonpolar bulk — in contrast to the large, flexible, or heteroaromatic-rich scaffolds that scored well throughout the dataset.

DISCUSSION AND CONCLUSION

Interpretation of the HIV-1 Protease Inhibitor Signal

This screen's central finding — that all six HIV-1 protease inhibitors in the library scored among the strongest binders to BACE1's active site, occupying five of the top eight overall ranks — has a clear mechanistic rationale. BACE1 and HIV-1 protease are both aspartic proteases that share a conserved catalytic mechanism built around a dyad of active-site aspartate residues [18]. BACE1 inhibitor medicinal chemistry has historically drawn directly on design principles developed for HIV protease and renin inhibitors, on the premise that transition-state-mimicking isosteres effective against one family member should, in principle, transfer to others [18, 19]. This is not merely a theoretical argument: HIV-1 protease inhibitors ritonavir and lopinavir have been experimentally co-crystallized with, and shown to inhibit, the unrelated malarial aspartic proteases plasmepsin II and X [20], and an independent 2025 multitarget virtual screen of approved drugs against Alzheimer's disease targets identified the HIV protease inhibitor saquinavir as a top-ranking hit [21], a finding this study's independent screen corroborates and substantially extends by testing all six clinically available HIV-1 protease inhibitors rather than one.

The consistency of this result across six independently developed drugs is itself informative. Lopinavir, saquinavir, darunavir, nelfinavir, atazanavir, and ritonavir were designed by different pharmaceutical companies over roughly two decades, sharing only their target (HIV-1 protease) and, consequently, their general peptidomimetic, transition-state-mimicking pharmacophore. That all six nonetheless converge on strong, catalytically-engaged docking scores against BACE1 — rather than a subset showing activity while others score randomly — argues against the result being an artifact of any single compound's idiosyncratic structure, and is more consistent with a genuine, shared structure–activity relationship rooted in the conserved aspartic-protease catalytic geometry.

This finding should be weighed against the well-documented history of BACE1 inhibitor clinical failure. Verubecestat, lanabecestat, atabecestat, umibecestat, and elenbecestat all achieved potent target engagement in humans, yet none produced clinical benefit, and several caused cognitive worsening [8–12]. This has led some researchers to argue that the field's fully-blocking, high-potency inhibitors may have been mechanistically miscalibrated, and that partial or differently-timed BACE1 modulation could be a safer therapeutic window [13]. HIV-1 protease inhibitors, notably, are not optimized for full, irreversible BACE1 blockade — they were selected by this screen's method to bind well computationally, not for any particular functional outcome at BACE1 — and whether their predicted binding would translate to partial, modulatory, or fully inhibitory functional activity at BACE1 is an open, experimentally answerable question raised directly by these results.

An Unexplained Secondary Hit: Montelukast

Montelukast's strong score (−9.0 kcal/mol, the best non-protease-inhibitor result) is a genuine outlier that does not fit either the peptidomimetic or heteroaromatic-sulfonamide pattern observed elsewhere in the dataset. As a large, rigid leukotriene-receptor antagonist bearing a chlorquinoline, cyclopropyl-carboxylic acid, and biphenyl system, its favorable score is most plausibly attributable to extensive shape complementarity with the pocket across many subpockets simultaneously — a distinct route to high affinity from either the protease inhibitors' transition-state mimicry or the smaller heteroaromatic NSAIDs' polar contacts. This finding is reported as an unexplained but robust result meriting independent follow-up, rather than forced into either of the other structural narratives.

The clean gradient observed within the eight-compound NSAID subgroup — tracking heteroaromatic and sulfonamide content rather than pharmacological subclass or molecular size alone — and the consistently weak performance of small polar nucleoside analogs and rigid cage amines together suggest that shape and polarity complementarity to the pocket's specific geometry, rather than any single pharmacological mechanism or drug class label, is the primary determinant of predicted binding affinity in this system. This is consistent with the broader medicinal-chemistry principle that a binding pocket's local geometry, not a compound's therapeutic indication, ultimately governs molecular recognition.

Limitations

Several limitations bear directly on how these results should be interpreted. First, and most importantly, AutoDock Vina and other empirical scoring functions used by CB-Dock2 are only weak-to-moderate quantitative predictors of true experimental binding affinity; docking scores should be understood as a hypothesis-generating rank-ordering tool, not a measurement of binding free energy [24]. Blind virtual screens of this kind routinely produce both false positives (favorable scores for non-binders) and false negatives (poor scores for true binders), and standard practice in the field is to treat a single-method docking score as a starting hypothesis requiring orthogonal validation — exactly the role the atabecostat re-docking benchmark was intended to serve in this study.

Second, this screen used a single static AlphaFold-predicted conformation of BACE1 and CB-Dock2's automated cavity-detection algorithm (CurPocket), which depends on the quality of a single geometric cavity detector; BACE1's flap loop is known to be conformationally flexible in solution, and a rigid-receptor docking protocol cannot capture induced-fit binding effects that could meaningfully change predicted affinities, particularly for the larger, more flexible protease-inhibitor ligands. Third, because this was a purely computational screen, no experimental confirmation — enzymatic inhibition assay, thermal shift assay, or structural (crystallographic/cryo-EM) validation — has yet been performed on any candidate; docking results alone cannot distinguish a true binder from a compound that merely fits the pocket's shape without productively engaging the catalytic mechanism. Fourth, HIV-1 protease inhibitors are generally known to have limited blood–brain-barrier penetration, a pharmacokinetic property this purely structure-based screen does not account for and which would need to be addressed before any of these compounds could be considered a realistic central-nervous-system

repurposing candidate. Finally, at least one published medicinal chemistry report describes an HIV protease inhibitor-derived compound found to be inactive against BACE1 in enzymatic assay, indicating that structural plausibility of cross-reactivity within the aspartic protease family does not guarantee potent inhibition for every compound in this class — a caveat this study's purely computational method cannot resolve.

Conclusion

This 36-compound blind docking screen of FDA-approved anti-inflammatory and antiviral drugs against a validated, literature-cross-checked model of the BACE1 active site identified a consistent, mechanistically rationalized signal: all six HIV-1 protease inhibitors screened ranked among the strongest predicted binders, occupying five of the top eight positions overall and meeting or approaching a literature benchmark score for a clinically advanced BACE1 inhibitor. This result is consistent with the shared aspartic-protease catalytic mechanism linking BACE1 to HIV-1 protease and with prior, independent computational and experimental evidence of cross-family aspartic-protease inhibition. Montelukast emerged as a notable secondary hit through an apparently distinct structural mechanism, and a clear structure–activity gradient was observed within the NSAID subclass. These findings nominate HIV-1 protease inhibitors, and more specifically their shared peptidomimetic, transition-state-mimicking pharmacophore, as a testable, structurally grounded hypothesis for BACE1-directed drug repurposing — while underscoring, consistent with standard practice in computational drug discovery, that these results represent a starting hypothesis for experimental follow-up rather than proof of therapeutic activity.

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docking runs. The author carefully reviewed, verified, and takes full responsibility for the accuracy, originality, and integrity of the published work.

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TABLES

Table 1. BACE1 active-site residues identified from the literature, shown in both classic (mature-protein) and full-precursor (AlphaFold/UniProt) numbering. A consistent +61-residue offset was applied across all reported residues.

Feature	Classic (mature-protein) numbering	Full-precursor / AlphaFold numbering
Catalytic dyad	Asp32, Asp228	Asp93, Asp289
Flap loop	Tyr71, Thr72, Gln73	Tyr132, Thr133, Gln134
S1 pocket	Leu30, Phe108	Leu91, Phe169
S1' pocket	Gly34	Gly95
S2 pocket	Thr231, Thr232	Thr292, Thr293

Table 2. Complete ranked docking results for all 36 screened compounds, sorted from strongest (most negative) to weakest predicted binding affinity. All compounds docked to the same catalytic-site cavity.

Ran k	Compound	Drug Class	Vina Score (kcal/mol)
1	Lopinavir	antiviral (HIV protease inhibitor)	-10.6
2	Saquinavir	antiviral (HIV protease inhibitor)	-10.1
3	Darunavir	antiviral (HIV protease inhibitor)	-9.6
4	Nelfinavir	antiviral (HIV protease inhibitor)	-9.5
5	Atazanavir	antiviral (HIV protease inhibitor)	-9.2
6	Montelukast	anti-inflammatory (leukotriene receptor antagonist)	-9.0
7	Baricitinib	anti-inflammatory (JAK inhibitor)	-8.9
8	Ritonavir	antiviral (HIV protease inhibitor)	-8.9
9	Celecoxib	NSAID (COX-2 selective)	-8.8
10	Methotrexate	anti-inflammatory (DMARD)	-8.6
11	Piroxicam	NSAID	-8.5
12	Sulindac	NSAID	-8.5
13	Dexamethasone	anti-inflammatory (corticosteroid)	-8.5
14	Sulfasalazine	anti-inflammatory (DMARD)	-8.4
15	Prednisone	anti-inflammatory (corticosteroid)	-8.4
16	Meloxicam	NSAID (COX-2 preferential)	-8.3
17	Apremilast	anti-inflammatory (PDE4 inhibitor)	-8.3
18	Tofacitinib	anti-inflammatory (JAK inhibitor)	-8.0
19	Indomethacin	NSAID	-7.9
20	Entecavir	antiviral (nucleoside analog)	-7.7
21	Diclofenac	NSAID	-7.6
22	Ketoprofen	NSAID	-7.6
23	Leflunomide	anti-inflammatory (DMARD)	-7.6
24	Naproxen	NSAID	-7.5
25	Valacyclovir	antiviral (nucleoside analog prodrug)	-7.0
26	Zidovudine	antiviral (nucleoside RT inhibitor)	-7.0
27	Hydroxychloroquine	anti-inflammatory (antimalarial/DMARD)	-6.9
28	Oseltamivir	antiviral (neuraminidase inhibitor)	-6.9
29	Tenofovir	antiviral (nucleotide RT inhibitor)	-6.8
30	Ibuprofen	NSAID	-6.6
31	Acyclovir	antiviral (nucleoside analog)	-6.3
32	Ribavirin	antiviral (nucleoside analog)	-6.3
33	Rimantadine	antiviral (M2 ion channel blocker)	-6.3
34	Aspirin	NSAID	-6.2
35	Amantadine	antiviral (M2 ion channel blocker)	-6.1
36	Lamivudine	antiviral (nucleoside RT inhibitor)	-6.1